

Extreme Value Theory and Information Dynamics of Seismicity in Central-Southern Italy

Personal project for the Advanced Statistics exam

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Abstract

In this work, I applied statistical tools to an earthquake catalog retrieved from the Istituto Nazionale di Geofisica e Vulcanologia (INGV) to analyze Central-Southern Italian seismicity. After a preliminary characterization of the dataset, I modeled the extreme value distribution using a Generalized Extreme Value (GEV) approach. The results yield a negative shape parameter ξ , albeit very close to zero, suggesting that the extreme value distribution falls within the basin of attraction of a Weibull distribution (upper bounded). In the second part of the study, I focus on a dynamical analysis of the 2016-2017 Central Italy seismic swarm. Specifically, I investigate the temporal evolution of Shannon Entropy and Fisher Information before, during, and after the sequence. With this analysis, I aim to detect correlations between mainshocks and aftershocks, highlighting deviations from linearity and the violation of the Poisson process assumption typically associated with random seismic events.

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1 Introduction and Dataset Characterization

1.1 Dataset description: Central-Southern Italy Earthquake Catalog

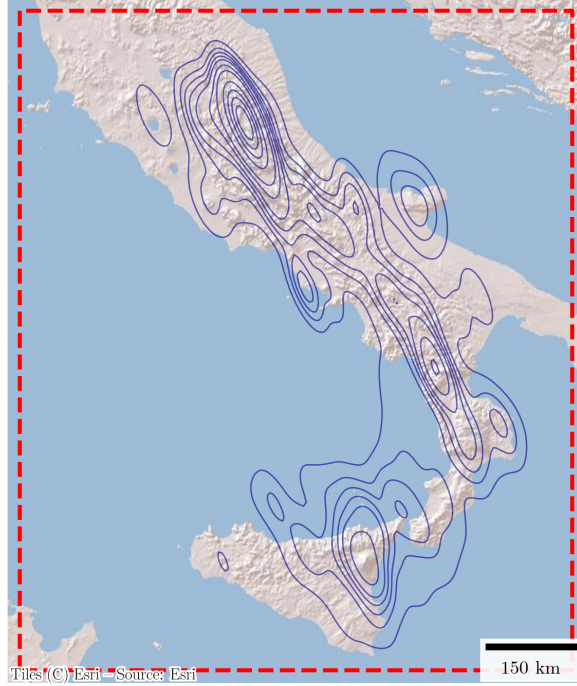


Figure 1: The map shows the level curve density of the events in the raw dataset. The dashed red box represents the boundaries of the analyzed spatial region.

The analysis presented below is based on an earthquake catalog retrieved from the INGV website [Istituto Nazionale di Geofisica e Vulcanologia]. The dataset covers the period from January 1985 to November 2025 within the spatial region defined by the coordinates: latitude 36.5° – 43.8° and longitude 10.0° – 18.0° , as shown in Figure 1. The raw catalog comprises 211,059 events, characterized by magnitudes in the range $0.1 \leq m \leq 6.5$ and depths of $0.1 \leq d \leq 644.4$ km.

1.2 Data Pre-processing

To ensure the statistical reliability of the dataset, I adopted a completeness magnitude threshold of $M_c = 2.5$. I identified this value as the magnitude below which the observed data deviate

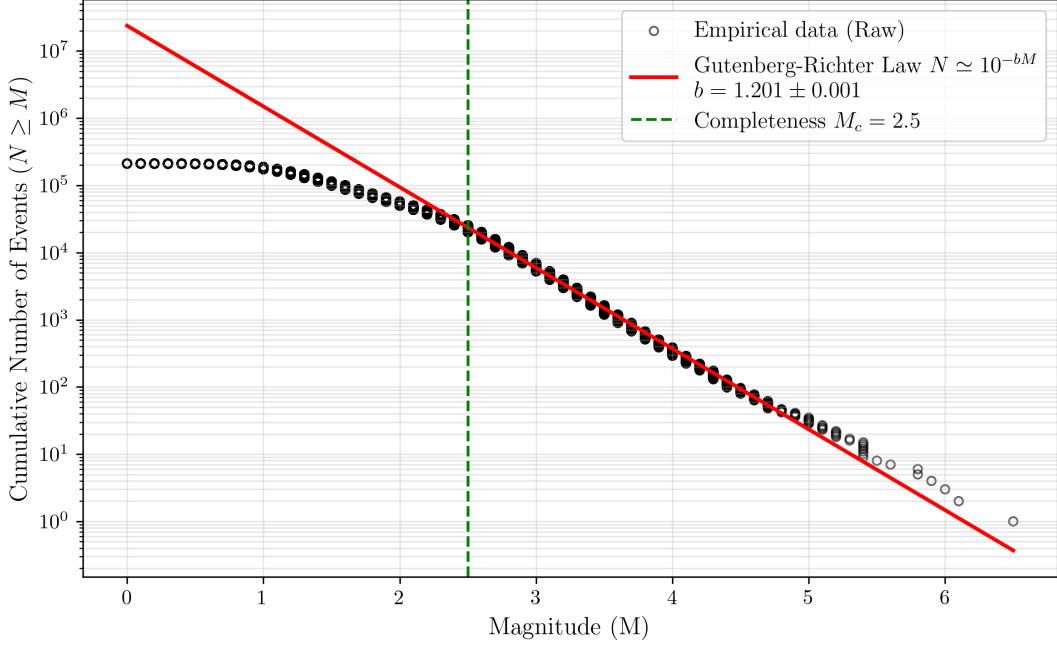


Figure 2: Gutenberg-Richter law fit. The data (scatter points) which don't follow the theoretical curve (solid red line) are discarded for the forward analysis. The dashed green line indicates the completeness threshold, $M_c = 2.5$.

from the linear trend predicted by the Gutenberg-Richter law [Mignan and Woessner, 2012] on a log-linear scale, as shown in Figure 2. This departure from linearity is attributed to catalog incompleteness caused by the limited detection sensitivity of the seismic network for smaller events. Consequently, all events with magnitude $m < M_c$ were discarded. Notably, this statistical threshold coincides with the perceptibility limit defined by INGV, marking the threshold for events generally felt by the population. The resulting filtered catalog contains 25,526 events; their cumulative count over time is displayed in Figure 3. To avoid mixing distinct seismic populations, which could distort the statistical analysis, I first divided the dataset into two subsets based on depth: a shallow earthquake catalog ($d \leq 70$ km) comprising 23,232 events, and a deep earthquake catalog ($d > 70$ km) containing 2,294 events. This segmentation aligns with the standard classification adopted by the INGV, which defines seismicity with $d \leq 70$ km as shallow, distinguishing it from deeper events.

However, raw seismic catalogs typically exhibit strong spatiotemporal correlations due to the

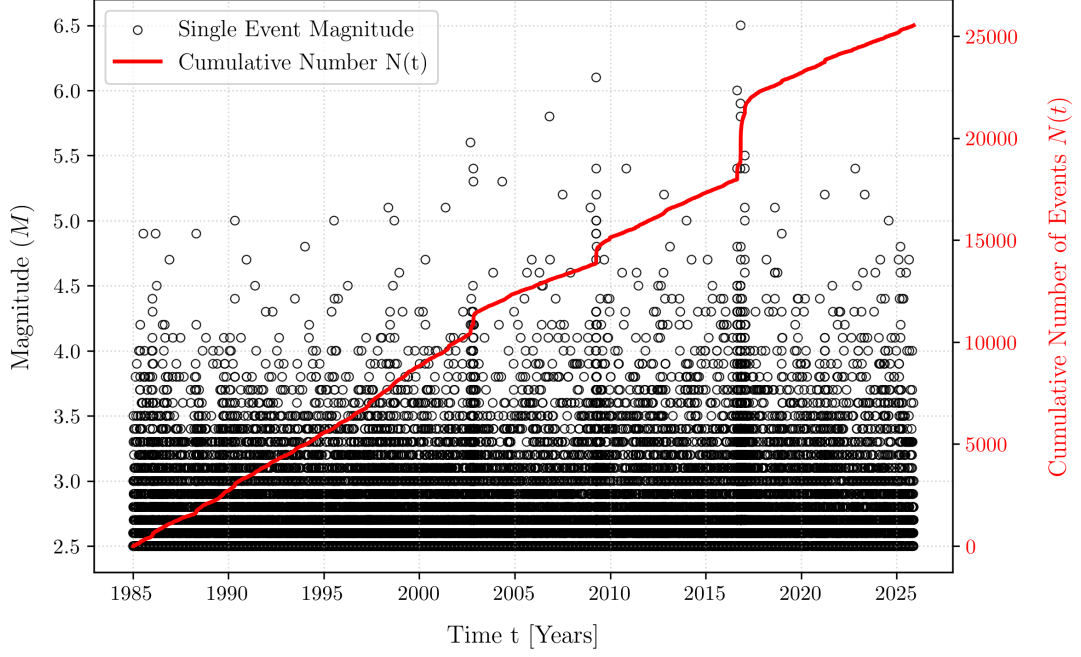


Figure 3: Observed event in the catalog (1985-2025). Only the events of magnitude $M \geq 2.5$ are shown. The red line indicates the cumulative number of events.

presence of mainshocks, foreshocks, and aftershocks. Since the subsequent statistical analysis (Extreme Value Theory) requires the assumption of independent and identically distributed (i.i.d.) variables, a declustering procedure was necessary to remove dependent events. I applied the windowing algorithm proposed by Gardner and Knopoff (1974) [van Stiphout et al., 2012], using the time and distance windows specified in Appendix A. This process effectively removed the aftershock sequences, resulting in a final dataset of 9,171 shallow mainshocks and 1,827 deep mainshocks, which constitute the basis for the hazard analysis presented in the following sections. The spatial distributions of these subsets are illustrated in Figure 4.

2 Modeling Extremes: The EVT Approach

In this section, I analyze the seismic hazard by focusing on the statistics of extreme events, specifically by studying the properties of the block maxima distributions for both the shallow and deep earthquake catalogs. According to the Fisher-Tippett-Gnedenko theorem, the distribution

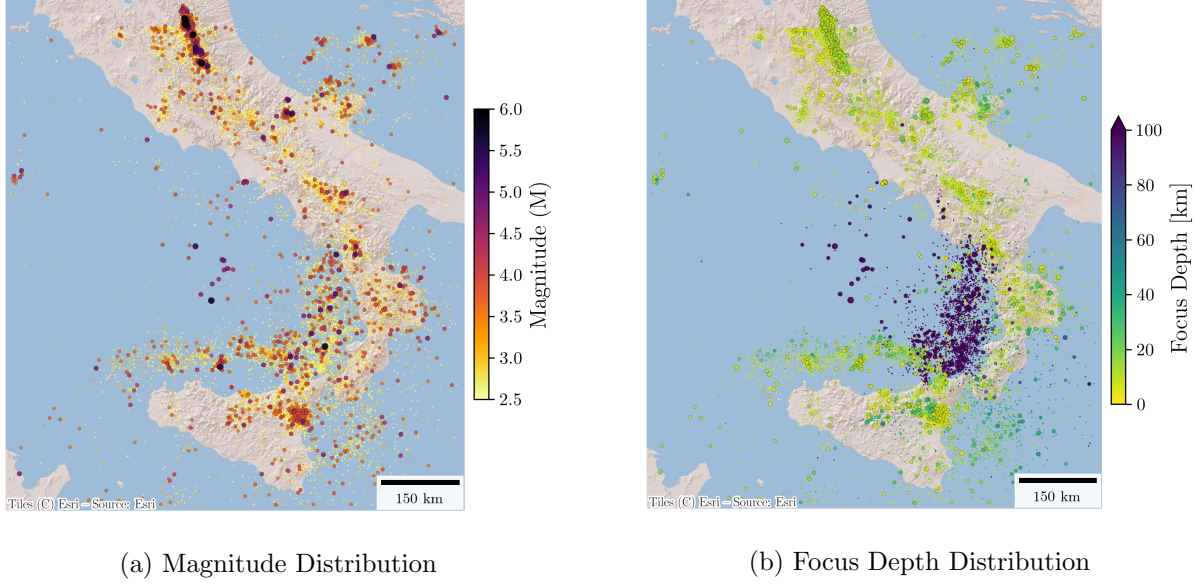


Figure 4: Spatial distribution of the Central-Southern Italy seismicity. (a) Geographical distribution colored by magnitude. (b) Geographical distribution colored by hypocentral depth.

of block maxima converges asymptotically to the Generalized Extreme Value (GEV) distribution. The cumulative distribution function (CDF) of the GEV is given by:

$$G(x; \mu, \sigma, \xi) = \exp \left\{ - \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-1/\xi} \right\} \quad (1)$$

with $-\infty < \mu < \infty$, $\sigma > 0$ and $-\infty < \xi < \infty$, and defined for $\{x : 1 + \xi(x - \mu)/\sigma > 0\}$, where μ , σ , and ξ are the location, the scale, and the shape parameters, respectively. The shape parameter ξ is of particular physical significance as it governs the tail behavior of the distribution. In this study, I adopt the standard sign convention:

- $\xi > 0$: Corresponds to the Fréchet distribution. This implies a “heavy-tailed” distribution with no finite upper limit, suggesting that magnitudes can theoretically increase indefinitely (though with decreasing probability).
- $\xi < 0$: Corresponds to the Weibull distribution. This implies a “short-tailed” or bounded distribution, suggesting the existence of a physical upper limit (M_{max}) to the magnitude.

- $\xi \rightarrow 0$: Corresponds to the Gumbel distribution, characterized by an exponential light tail.

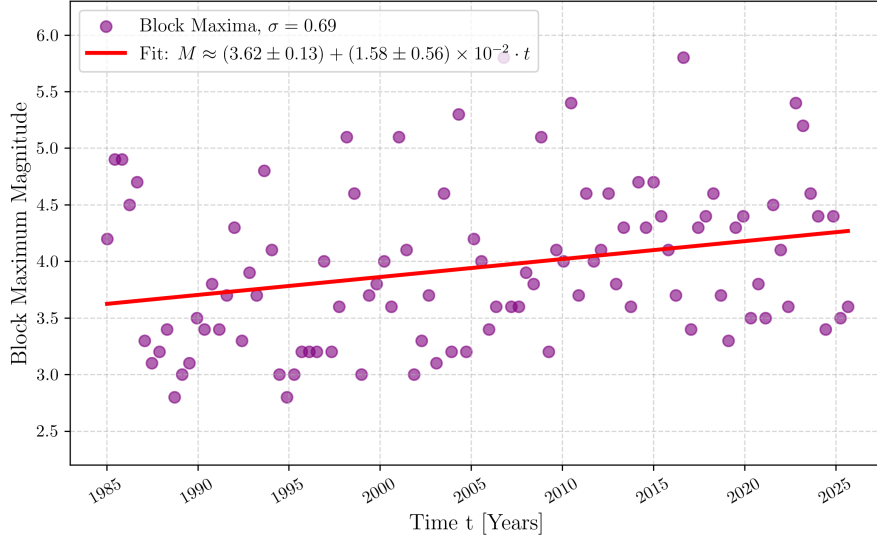
By performing a statistical analysis to model the empirical maxima distributions (GEV fit), I will determine the most appropriate domain of attraction for the shallow and deep catalogs. Finally, I will use the resulting GEV models to estimate seismic hazard metrics, specifically the return periods of extreme events and the risk probability over fixed time horizons. In the subsequent graphical analysis (including CDF comparisons, Q-Q plots, and Return Period curves), the empirical cumulative probabilities of the observed block maxima are estimated using the Gringorten plotting position [Gringorten, 1963]:

$$P_i = \frac{i - 0.44}{N + 0.12} \quad (2)$$

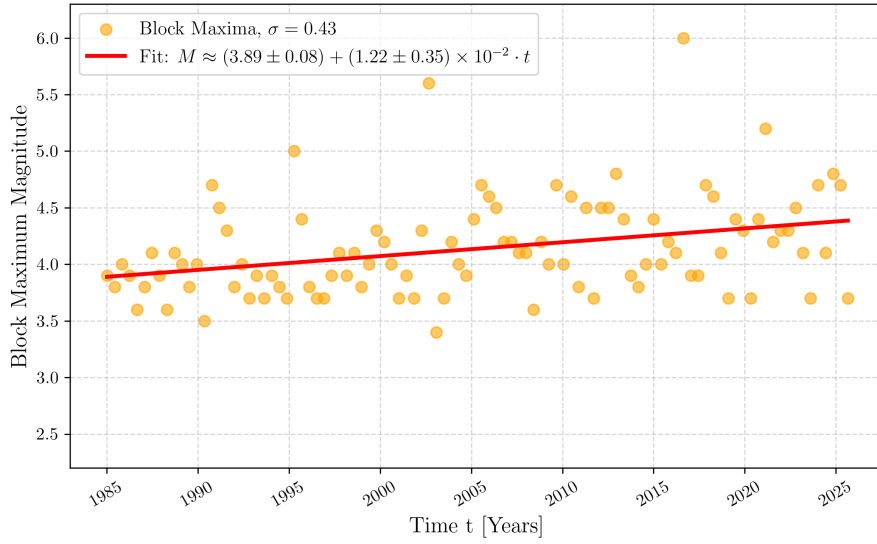
where i denotes the rank of the observed maximum (sorted in ascending order) and N is the total number of block maxima. This estimator is preferred over standard rank-based approaches (such as $P_i = i/N$ or $P_i = i/(N + 1)$) because it avoids singularities (e.g., $\log(0)$) in logarithmic transformations. Furthermore, Gringorten originally derived this expression specifically for double-exponential (Gumbel) distributions to minimize bias in the upper tail. Since the GEV distribution converges to the Gumbel form when the shape parameter $\xi \rightarrow 0$, this plotting position remains a robust and widely adopted choice for geophysical applications, where the shape parameter is expected to be close to zero.

2.1 The Block Maxima Method

I derived the maxima distributions from the two catalogs using the Block Maxima Method. Specifically, I divided the magnitude time series into N temporal windows of fixed duration and computed the maximum magnitude for each interval. These maxima were then collected to obtain the distribution, which I analyzed using Extreme Value Theory (EVT). It is crucial to emphasize that for EVT to be valid, the variables must be independent and identically distributed (i.i.d.). Post-declustering, I proceeded with the analysis under the assumption of stationarity. Although the scatter plots in Figure 5 suggest a slight positive linear trend (with a slope of approximately 10^{-2} per year), I consider this variation negligible for the purposes



(a) Deep Seismicity



(b) Shallow Seismicity

Figure 5: Temporal evolution of block maxima magnitudes (M_{max}) for (a) deep and (b) shallow seismicity. Red lines represent linear trend fits, with t indicating years since 1985. The standard deviation (σ) is provided in the legends to contextually compare the inherent variability of the maxima with the observed linear drift.

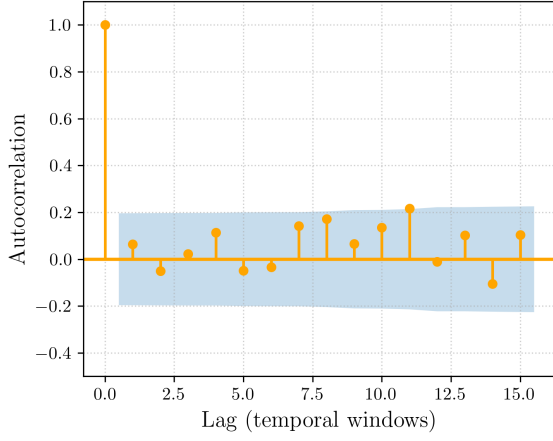
of this study. My quantitative analysis reveals that the total drift observed over the analyzed period is comparable to the standard deviation of the data ($m_{drift} \simeq 1\sigma$ for both datasets). Consequently, I treated the systematic trend as effectively indistinguishable from the inherent stochastic noise of the distributions. From a geophysical perspective, a genuine change in the seismogenic process over such a short timescale (40 years) is unlikely. It is more plausible that this trend is an artifact resulting from improvements in detection technology or the occurrence of specific high-energy sequences, such as L'Aquila 2009 or Amatrice 2016, that temporarily elevated the moving average. To rigorously validate the stationarity assumption, I performed a standard Dickey-Fuller (DF) test on both datasets. The results, summarized in Table 1, provide evidence against the null hypothesis of non-stationarity (unit root) at the 5% significance level ($\alpha = 0.05$). For both deep and shallow mainshocks, the p-values are several orders of magnitude smaller than the threshold ($p \ll 0.05$), confirming that the series are strongly mean-reverting.

Table 1: Results of the Dickey-Fuller stationarity test (lag=1) performed on the block maxima series. The significance level was set at $\alpha = 0.05$.

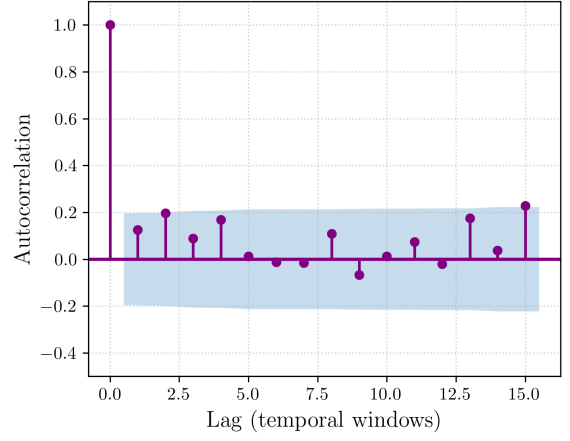
Dataset	Test Statistic	p-value	Result
Deep Earthquakes	-5.41	3.26×10^{-6}	Stationary
Shallow Earthquakes	-7.09	4.54×10^{-10}	Stationary

Therefore, I treated the series as stationary without introducing significant bias¹. Regarding the independence of the maxima, I performed autocorrelation measurements to guarantee that the values are independent, as shown in Figure 6. I selected the fixed window size after testing the temporal interval of stability, i.e. the duration for which the estimate of the shape parameter ξ remains approximately constant. The criteria ensuring stability for the statistical tests are twofold: first, ensuring a sufficient number of events within each block (an average of at least 8) so that the block maximum represents a true extreme; second, ensuring a sufficient number of blocks (at least 100 maxima) to perform a reliable GEV fit. I identified a suitable compromise

¹ I also validated this assumption by performing a sensitivity analysis on detrended data, which yielded GEV parameters (ξ, σ) compatible with those of the original series within their respective uncertainty ranges.



(a) Shallow Seismicity



(b) Deep Seismicity

Figure 6: Autocorrelation Function (ACF) plots of the block maxima series. The points falling within the shaded blue area (95% confidence interval) indicate that there is no significant serial correlation between consecutive windows.

based on the catalog density λ , defined as:

$$\lambda = \frac{n_{events}}{T_{days}} \quad (3)$$

The objective is to satisfy the condition:

$$\lambda t_{window} \gg 1 \quad (4)$$

For my catalogs, the densities are:

$$\begin{aligned} \lambda_{shallow} &\simeq 0.61 \\ \lambda_{deep} &\simeq 0.12 \end{aligned} \quad (5)$$

and the total time period is $T_{days} \approx 14,934$ days. Consequently, I selected a window of $t_{window} = 150$ days as a suitable compromise, yielding 100 block maxima per catalog and ensuring sufficient average density within each block. However, I observed that the dataset for deep earthquakes is relatively limited; as a result, the estimates for this subset exhibit higher instability compared to the shallow catalog results, which I consider more robust due to the larger sample size.

2.2 GEV Fit and Parameter Estimation

To estimate the parameters of the GEV distribution (location μ , scale σ , and shape ξ) I adopted the Generalized Maximum Likelihood (GML) method. As demonstrated in [Martins and Steidinger, 2000], GML estimators generally outperform standard Maximum Likelihood (MLE) and Method of Moments estimators, particularly for small sample sizes and when the shape parameter ξ is slightly negative (bounded distributions), as is typical of geophysical phenomena.

The GML method mitigates the instability of MLE in small samples by introducing a Bayesian prior on the shape parameter. In this analysis, I employed a specific geophysical prior modeled as a Beta distribution with parameters $p = 9$ and $q = 6$, defined over the interval $\xi \in [-0.5, 0.5]$. This prior choice restricts the estimation to a physically reasonable range, penalizing absurd values while allowing sufficient flexibility for the data to drive the fit.

To ensure the robustness of the parameter estimates, I implemented a Bootstrap procedure ($N_{boot} = 100$), based on a reshuffle of the original magnitude series. It is important to specify that the reshuffling algorithm was designed to preserve the original statistical properties of the catalog. The sequence of observed magnitude values was not modified (i.e., no resampling with replacement was performed on the magnitudes themselves). Instead, I assigned random occurrence times to the fixed set of recorded magnitudes. This approach maintains the original event frequency (λ) of the catalog while generating diverse, yet physically consistent, distributions of block maxima for each iteration. The final best estimate for the shape parameter ξ corresponds to the median of these bootstrap realizations.

Finally, to quantify the statistical uncertainty, I calculated the 90% Confidence Intervals (CI). Following the methodology described in [Pisarenko et al., 2014], I performed a Monte Carlo simulation by generating $N_{MC} = 500$ synthetic datasets based on parameters estimated via the Bootstrap procedure. The 90% confidence interval boundaries were determined using the 5th and 95th percentiles ($Q_5 - Q_{95}$) of the resulting distribution of simulated parameters. Additionally, I computed the Root Mean Square Error ($RMSE_{\xi}$) on the synthetic parameter distribution to assess the variability of the estimation.

The results of this analysis for both catalogs are summarized in Table 2. For both seismic populations, I obtained a negative shape parameter, implying that the respective distribution

Table 2: Estimates of the GEV location (μ), scale (σ), and shape (ξ) parameters. The reported values represent the median of the bootstrap distribution. For the shape parameter ξ , the 90% Monte Carlo confidence interval (Q_5 – Q_{95}) and the RMSE are also included. (Sign convention: $\xi < 0$ indicates a Weibull distribution)

Dataset	μ_{best}	σ_{best}	Q_5	ξ_{best}	Q_{95}	RMSE $_{\xi}$
Deep	3.66	0.51	-0.1489	-0.0312	0.0502	0.0640
Shallow	3.89	0.31	-0.1903	-0.0940	-0.0047	0.0569

of maxima is upper-bounded. This result aligns with seismological theory and physical constraints, as the Earth’s finite energy budget implies the existence of an M_{max} in every region. However, I did not estimate this specific value in the current analysis; the shape parameter’s proximity to zero, combined with the scarcity of data in the tails, would render such an estimation unstable. As shown in the table, the confidence interval for the shallow catalog lies entirely within the negative domain (albeit close to zero), reflecting the more robust statistics available for this dataset. Conversely, the confidence interval for the deep catalog crosses zero, meaning the Gumbel hypothesis cannot be rejected for this distribution. A significant observation concerns the estimation uncertainty relative to the parameter value itself. For the shallow catalog, the RMSE $_{\xi}$ is substantial, representing approximately 50% of the estimated ξ value. This uncertainty becomes even more pronounced for the deep catalog, where the RMSE $_{\xi}$ exceeds the absolute value of the shape parameter itself (RMSE $_{\xi} > |\xi|$). This high relative error is understandable given that the estimated shape parameters for both datasets oscillate closely around zero. In such conditions ($\xi \approx 0$), small fluctuations in the data or the simulation process can lead to relative errors that appear large compared to the parameter’s small magnitude, without necessarily invalidating the overall fit stability. The fact that both shape parameters are close to zero is not unexpected: seismic data generally follow the Gutenberg-Richter law, which describes an exponential relationship. My data, in particular, closely adhere to this law from $m \geq M_c = 2.5$ up to the tail of the distribution ($m \gtrsim 4.5$), as we saw in Figure 2. The limitation, more pronounced for deep earthquakes than for shallow ones, probably lies in the

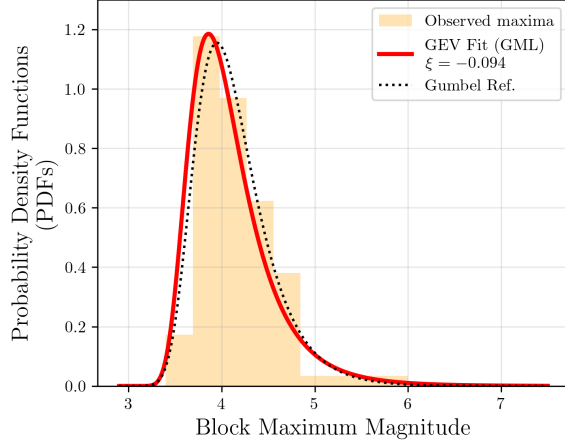
relatively short catalog duration (40 years). During this period, mainshocks with magnitude $m \gtrsim 4.5$ were few (17 shallow and 22 deep events). Consequently, the data were likely insufficient to fully characterize the curvature of the distribution's tail, meaning the well-sampled exponential portion of the data dominated the GEV fit, pulling the shape parameter towards the Gumbel limit ($\xi = 0$). Nevertheless, the GEV fit provides a satisfactory representation for both catalogs, as illustrated in Figure 7, which displays the empirical and modeled PDFs and CDFs. As previously noted, the fit appears more accurate for shallow earthquakes than for deep ones, with the latter exhibiting behavior more characteristic of a Gumbel distribution. This similarity is further highlighted by the Quantile-Quantile (Q-Q) plot in Figure 8. The plot demonstrates that data from both catalogs approximate the Gumbel distribution well; indeed, the regression analysis yielded an R^2 value of $\simeq 0.98$.

2.3 Risk Analysis: Return period and risk probability

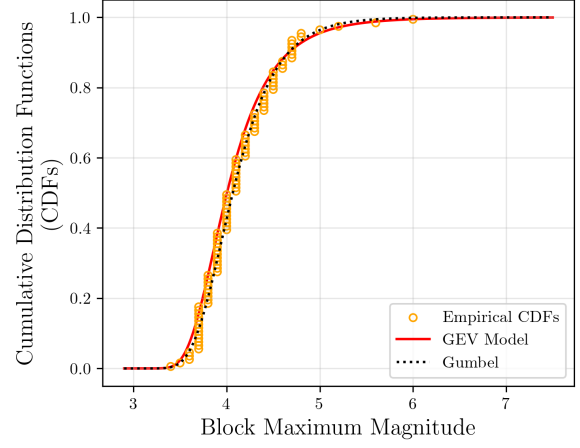
Building upon the previously discussed GEV fit, I performed a risk analysis to estimate the Return Period, τ_T , and the associated Exceedance Probability, p_t . These quantities are defined as follows:

$$\begin{aligned}\tau_T &= \frac{T}{1 - F(m)} \\ p_t &= 1 - [F(m)]^{N(t)}\end{aligned}\tag{6}$$

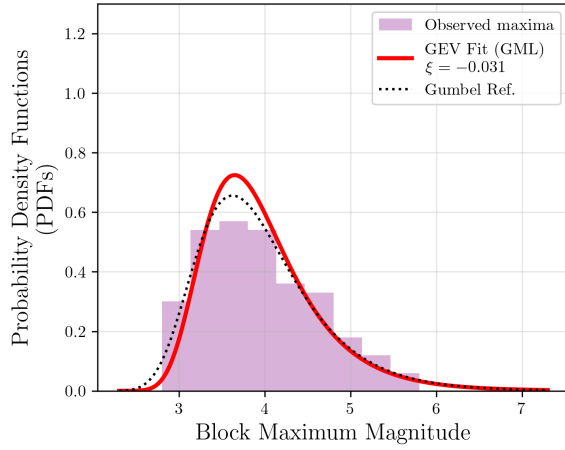
where T represents the duration of each block in years, $F(m) \equiv P(M \leq m)$ denotes the Cumulative Distribution Function (CDF), t represents the time horizon (in years) over which the risk is assessed, and $N(t)$ is the number of blocks, determined by the ratio of the time horizon t to the block duration T . The reliability of the GEV model is visually confirmed by the Return Period plots (Figures 9b and 9a). The empirical Return Periods (depicted as circles) adhere closely to the theoretical model (red line) across the majority of the magnitude range for both catalogs. As anticipated, slight deviations are observable in the tail of the distribution ($M \gtrsim 5.5$); these fluctuations are attributable to the scarcity of historical mainshocks in the high-magnitude range, which makes the empirical estimation more sensitive to individual events. Regarding the risk analysis, the Exceedance Probability curves (Figures 9d and 9c) quantify the seismic hazard over distinct time horizons ($t = 5, 40, 80$ years). As expected, the probability of



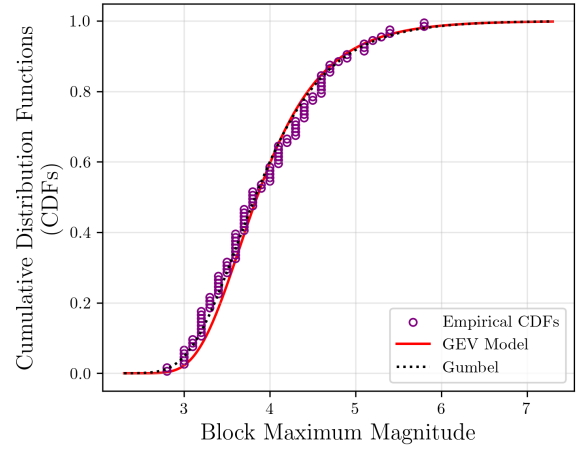
(a)



(b)



(c)



(d)

Figure 7: Comparison of maxima distributions for Shallow (top row) and Deep (bottom row) events. The left column displays the Probability Density Functions (PDFs), while the right column shows the empirical Cumulative Distribution Functions (CDFs). The GEV fits were derived from the original dataset.

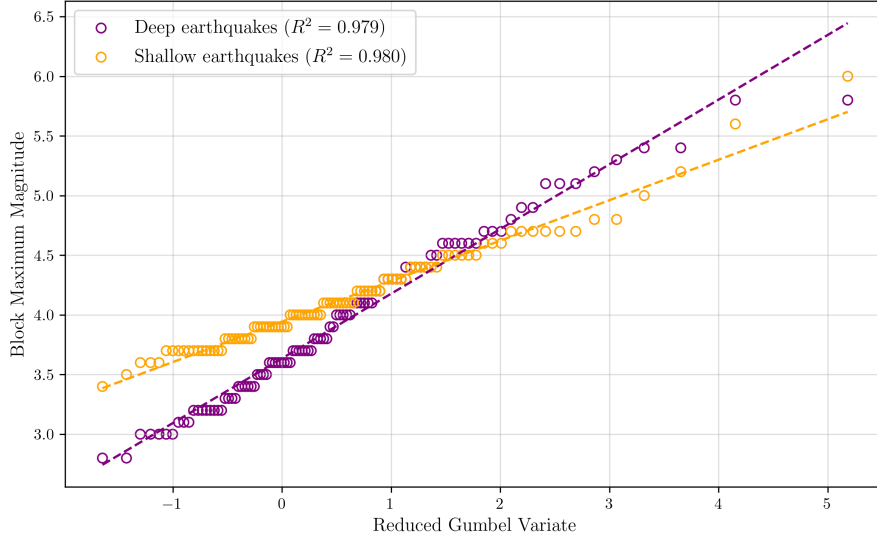
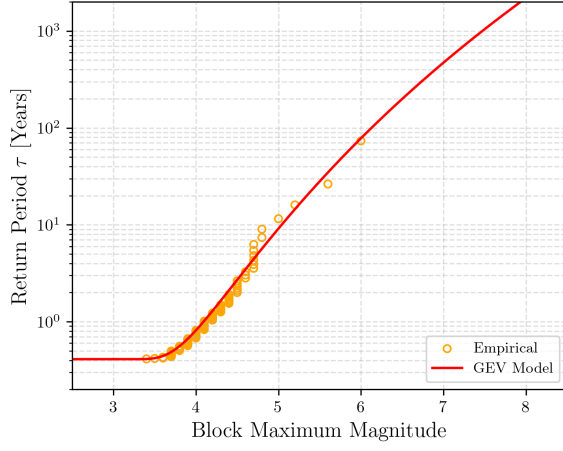
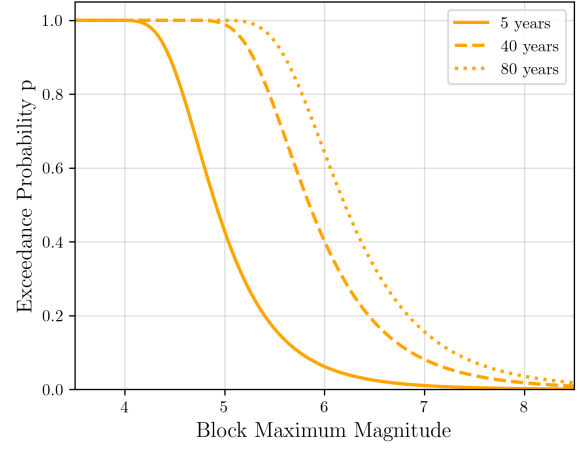


Figure 8: Quantile-Quantile (Q-Q) plot comparing the empirical data against the theoretical Gumbel distribution.

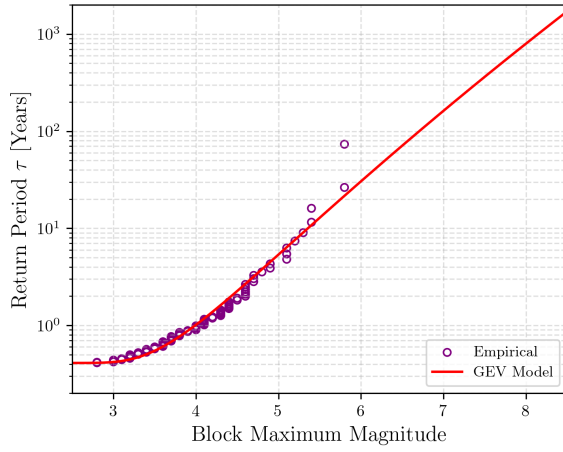
exceeding a given magnitude increases with the exposure time. It is worth noting a “saturation” effect for longer time horizons: for magnitudes $M \lesssim 5$, the curves for both 40 and 80 years converge towards unity ($p_t \approx 1$). This implies that, statistically, an event of such magnitude is almost certain to occur over a very long period (several decades). It is crucial to define the limits of validity for these long-term extrapolations. According to standard practices in extreme value analysis [Parrett, 1997, p.24], statistical reliability significantly decreases when the prediction horizon exceeds 2 times the duration of the observational catalog (in my case, $T_{obs} \approx 40$ years). Therefore, while the risk estimates for $t = 5$ and $t = 40$ years are supported by the temporal coverage of the data, the projections for $t = 80$ years (representing $2 \times T_{obs}$) should be interpreted with greater caution. Beyond this horizon, the uncertainty would likely dominate the results, potentially yielding physically unrealistic magnitude estimates due to the lack of constraints on the distribution’s upper tail, M_{max} .



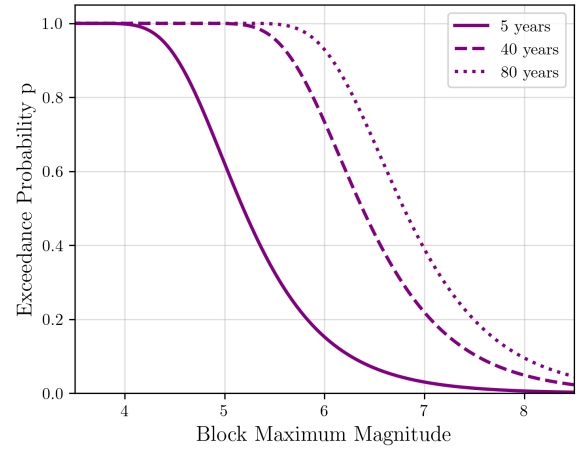
(a) Shallow Seismicity



(b) Shallow Seismicity



(c) Deep Seismicity



(d) Deep Seismicity

Figure 9: Return Period (left panels) and Risk probability (right panels) estimation for Shallow (upper panels) and Deep (lower panels) earthquakes. The estimations were made using GEV model.

3 Seismic Dynamics: An Information Theory Approach

In this section, I present a dynamic analysis of the seismic system's evolution, with a specific focus on the Amatrice-Visso-Norcia seismic swarm (2016-2017).

3.1 Application to the Amatrice Earthquake Swarm

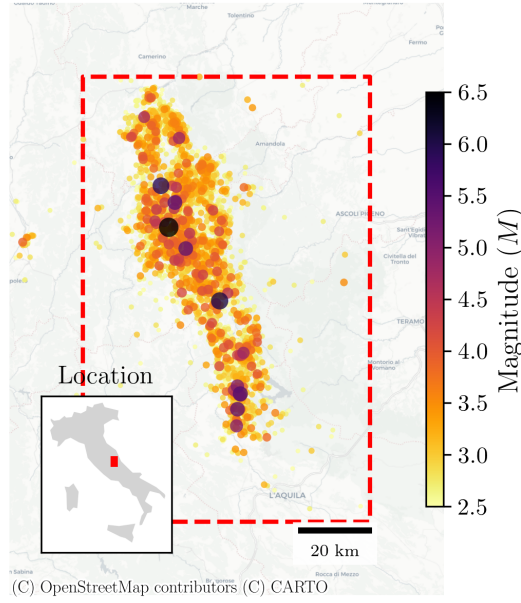


Figure 10: Spatial distribution of earthquakes sorted by magnitude (Central Italy seismic swarm, 2016-2017).

The analyzed dataset covers the period from January 2015 to December 2017 within the spatial region defined by the coordinates: latitude 42.32° - 43.09° and longitude 6.49° - 13.57° , as illustrated in Figure 10. I considered a catalog comprising 4,008 events, characterized by magnitudes in the range $2.5 \leq m \leq 6.5$ and depths of $0.5 \leq d \leq 25$ km. Consistent with my previous analyses, I excluded all events below the completeness magnitude ($m \leq M_c = 2.5$). Notably, I did not decluster the catalog, as my primary objective is to investigate the correlations between events. The four main-shocks of this sequence (Amatrice, Visso, Norcia, and Campotosto) represent the highest magnitude events in the catalog. As observed in Figure 3, the cumulative event count

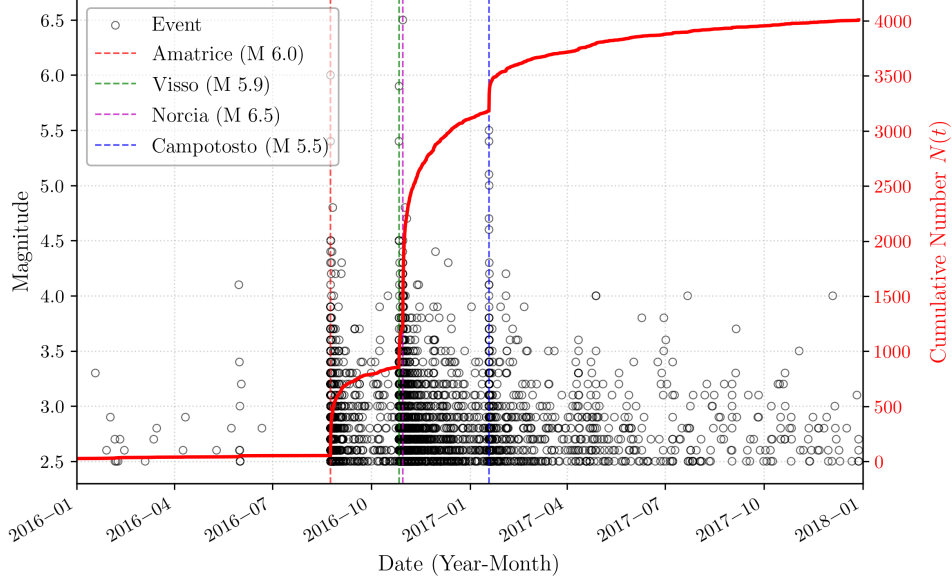


Figure 11: Cumulative number of events over time.

exhibits step-like behavior at points corresponding to these major events, breaking the linearity of the relation $N(t)$ vs t . Notably, the most significant step occurs during the 2016–2017 period, which coincides with the swarm activity. Figure 11 details the cumulative number of events for this specific period, clearly illustrating how the discontinuities align with the four mainshocks. This behavior suggests that a large event (mainshock) triggers numerous secondary events (aftershocks). Consequently, the assumption of a Poisson process is violated, as the events within the swarm exhibit strong mutual correlations. In this section, I aim to analyze these correlations using Information Theory tools (specifically Shannon Entropy and Fisher Information) to determine whether the presence of a swarm alters the system’s memory.

3.2 The Fisher-Shannon Information Plane

In this subsection I investigated the dynamic evolution of the seismic system by analyzing the relationship between two fundamental Information Theory metrics computed on the inter-event times [Telesca et al., 2004]: the Shannon Entropy, H_δ , and the Fisher Information, I_δ ,

respectively defined as²:

$$H_\delta = - \sum_{i=1}^{N_{bins}} p(\delta_i) \log p(\delta_i) \quad (7)$$

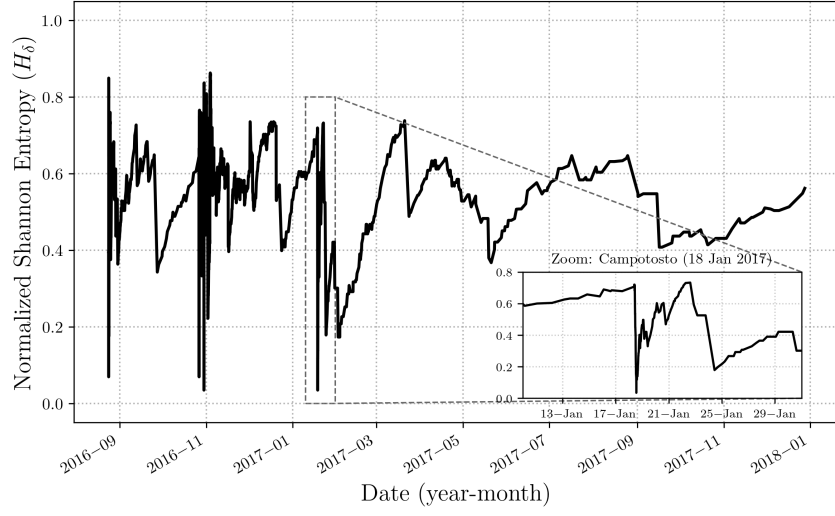
$$I_\delta = 4 \sum_{i=1}^{N_{bins}-1} \left(\sqrt{p(\delta_{i+1})} - \sqrt{p(\delta_i)} \right)^2 \quad (8)$$

where δ represents the series of inter-event times ($\delta_k = t_k - t_{k-1}$), $p(\delta_i)$ is the probability associated with the i -th bin of the probability density function (PDF), and N_{bins} is the number of bins used for the discretization. To capture the system's evolution, I applied a sliding window approach to the catalog using a size of $w = 100$ events. I estimated the PDF for each window via histograms with $N_{bins} = 5$; this configuration ensures an average of 20 events per bin, effectively minimizing estimation bias and guaranteeing the robustness of both metrics. Shannon Entropy is a measure of the disorder of the system; higher values of H_δ indicate a higher degree of randomness, typical of a Poissonian process. Conversely, Fisher Information (I_δ) is a local measure that quantifies the “sharpness” of the distribution and its internal organization. The temporal evolution of these metrics is presented in Figure 12. From this comparison, I observed a clear inverse correlation between the two indicators. During inter-seismic periods characterized by background activity far from the highlighted mainshocks, Shannon Entropy remains high while Fisher Information is low, indicating a disordered state resembling a random memoryless process. Conversely, in correspondence with the mainshocks (Amatrice, Visso, Norcia, and Campotosto), I detected dramatic drops in entropy synchronized with sharp spikes in Fisher Information. Physically, these results suggest that the mainshocks induce a strong internal organization within the fault system, breaking the stochastic nature of the background seismicity.

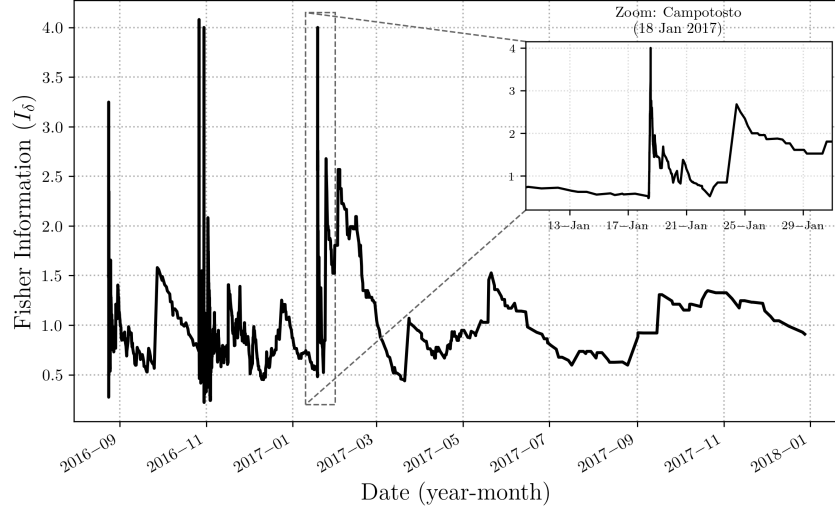
3.3 Trajectory in the Fisher-Shannon Information Plane

To visualize the global dynamics of this phase transition, I projected the system's evolution onto the Fisher-Shannon Information Plane (FSIP). In this representation, I plotted the Shannon Entropy Power ($N_\delta = e^{2H_\delta}$) against the Fisher Information (I_δ). The dashed line in Figure

² These definitions refer to the discrete formulation.



(a) Normalized Shannon Entropy (H_δ)



(b) Fisher Information (I_δ)

Figure 12: Comparison between (a) Shannon Entropy and (b) Fisher Information. Note the inverse correlation: peaks in Fisher Information correspond to sharp valleys in Shannon Entropy. From left to right the four peaks (valleys) correspond to the main events of the swarm: Amatrice ($M = 6.0$), Visso ($M = 5.9$), Norcia ($M = 6.5$) and Campotosto ($M = 5.5$). For both plots a zoom on the Campotosto event is included.

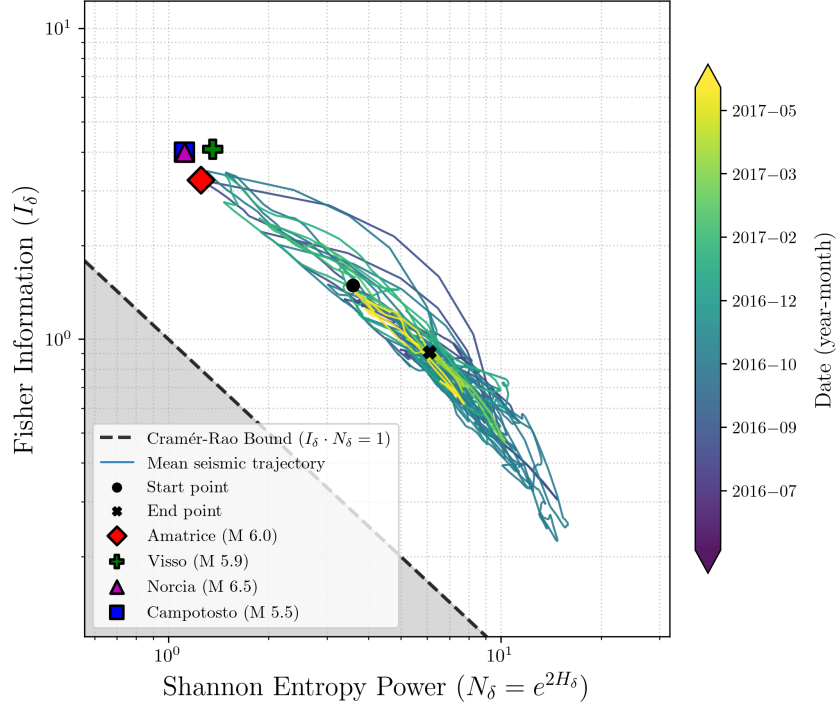


Figure 13: Evolution of the seismic system in the Fisher-Shannon Plane. The multicolor line represents the mean trajectory (smoothed), colored by time. The start (circle) and end (cross) markers indicate the initial and final states. The colored markers highlight the state of the system in proximity to the main-shocks. The dashed line represents the Gaussian limit.

13 represents the Cramér-Rao lower bound ($I \cdot N = 1$), which corresponds to the theoretical limit for Gaussian processes [Guignard et al., 2020]. The system’s trajectory reveals a complex dynamic loop. The continuous line represents the smoothed average state of the system (using a smoothing window $w_s = 5$). The mainshocks (colored markers) appear positioned significantly above the mean trajectory. This confirms that these events momentarily pull the system towards a state of maximum order (top-left corner), far exceeding the background dynamics. Throughout the sequence, the system remains distinct from the Cramér-Rao bound, confirming that the seismic swarm is a complex, non-linear process that never fully relaxes into a pure Gaussian noise state. Ideally, in a perfectly closed cycle, the system would return to its initial state. However, the start point (circle) is located at a higher Fisher Information level compared to the end point (cross). This discrepancy likely arises because the dataset begins approximately 1.5 years prior to the Amatrice earthquake; therefore, the “start” state likely intercepts a preparatory phase (foreshock activity) where the system is already organizing under stress. In contrast, the “end” point represents the tail of the relaxation phase, where the system progressively loses structure and settles into a state of higher disorder. In conclusion, the analysis demonstrates that the seismic swarm is a complex dynamical process where the system actively transitions from disorder to order during energy release, following a loop that alters its structural configuration.

4 Conclusions

In this work, I performed a comprehensive statistical analysis of Central-Southern Italian seismicity, combining Extreme Value Theory (EVT) for hazard assessment and Information Theory for dynamic characterization. The EVT analysis applied to the declustered catalog provided significant insights into the maximum magnitude distribution. The Generalized Extreme Value (GEV) fit yielded a shape parameter ξ very close to zero for both shallow and deep earthquakes, indicating a behavior that likely approximates the Gumbel distribution (exponential tail). However, the point estimates are consistently negative ($\xi \approx -0.09$ and $\xi \approx -0.03$), coherently with similar studies ([Pisarenko et al., 2010], [Pisarenko et al., 2014]). From a physical perspective, these negative values are consistent with the hypothesis that the magnitude distribution belongs to the Weibull domain of attraction, which implies the existence of a finite upper bound

(M_{max}) for seismic events. This interpretation is compatible with the concept of finite energy storage capacity of the fault systems, although for the deep distribution the statistical uncertainty (with confidence intervals crossing zero) prevents a definitive exclusion of the unbounded Gumbel case. As already discussed, this feature could be due to the scarcity of events on the tail. The associated risk analysis highlighted that high-magnitude events, while rare, have return periods that warrant careful consideration in long-term hazard planning. The dynamic analysis of the 2016–2017 seismic swarm revealed the complex temporal evolution of the source process. The Fisher-Shannon analysis demonstrated a clear inverse correlation between Shannon Entropy and Fisher Information. This behavior aligns with findings from similar studies ([Telesca et al., 2004], [Telesca et al., 2012]), confirming that seismic crises are characterized by a sharp increase in system order. Specifically, the system exhibits a transition from a disordered state during background activity to a highly organized state during mainshocks. The mainshocks act as impulsive singularities that maximize the system’s internal structure (I_{max}, H_{min}). The projection onto the Fisher-Shannon Information Plane revealed a loop-like behavior. Crucially, the system did not return to its exact initial state after the sequence; the end point showed higher entropy than the start point. This suggests that the seismic swarm likely altered the structural configuration of the fault network; it is also plausible that the initial state captured a preparatory foreshock phase. In conclusion, this study demonstrates that statistical physics provides a robust framework for decoding the complex dynamics of seismic systems. While I used Extreme Value Theory to investigate the probabilistic limits of energy release, Information Theory allowed me to characterize the system’s self-organization and its transition from stochasticity to ordered states.

A Gardner-Knopoff Declustering Algorithm

In this study, the declustering of the catalog was performed using the deterministic windowing method proposed by Gardner and Knopoff [van Stiphout et al., 2012]. This algorithm identifies sequences of aftershocks by applying spatial and temporal filters dependent on the magnitude of the main event. The fundamental assumption is that main-shocks follow a Poissonian distribution in time, while dependent events (aftershocks) cluster around them. For each event with magnitude M , the algorithm defines a spatial window $L(M)$ and a temporal window $T(M)$. Any subsequent event occurring within distance L and time T is flagged as an aftershock (or dependent event) and removed from the catalog to produce the background seismicity dataset.

A.1 Window Parameters

The dimensions of the windows scale with magnitude. The empirical relationships used to approximate the window boundaries are defined as follows:

$$L(M) = 10^{0.1238 \cdot M + 0.983} \quad [\text{km}] \quad (9)$$

$$T(M) = \begin{cases} 10^{0.5409 \cdot M - 0.547} & \text{if } M < 6.5 \\ 10^{0.032 \cdot M + 2.7389} & \text{if } M \geq 6.5 \end{cases} \quad [\text{days}] \quad (10)$$

Table 3 reports the discrete values originally proposed by Gardner and Knopoff (1974) for Southern California seismicity, which serve as the reference standard for the method.

Table 3: Window sizes for aftershock identification (Gardner and Knopoff, 1974).

Magnitude (M)	Distance L (km)	Time T (days)
2.5	19.5	6
3.0	22.5	11.5
3.5	26.0	22
4.0	30.0	42
4.5	35.0	83
5.0	40.0	155
5.5	47.0	290
6.0	54.0	510
6.5	61.0	790
7.0	70.0	915
7.5	81.0	960
8.0	94.0	985

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